Zayed Humayun

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Lecturer, Department of Computer Science and Engineering, BRAC University

Quantum computing, quantum learning theory, and theoretical computer science

EDUCATION

BRAC University, Dhaka, Bangladesh — Jun 2025 – present
MSc in Computer Science and Engineering

BRAC University, Dhaka, Bangladesh — Jan 2020 – Dec 2023
BSc in Computer Science

Selected coursework includes Quantum Computing, Randomized Algorithms, and Cryptography.

QUANTUM RESEARCH

Learning sparse pure states using single-qubit measurements — MSc thesis in progress

Developed an algorithm that reconstructs an n-qubit pure state with at most k nonzero amplitudes using single-qubit Pauli measurements. For arbitrary supports, the current analysis gives $\tilde{O}(k^2/\varepsilon)$ copies and $\tilde{O}(nk^2/\varepsilon)$ classical time for infidelity at most ε.

Adaptive Learning of Forest Graph States — 2026
MSc Network Science course project, BRAC University

Designed and simulated a protocol that chooses single-qubit X and Z measurements using outcomes from earlier copies. Investigated whether it can identify a forest with asymptotically fewer graph-state copies than fixed measurement strategies.

PREPRINTS

R. S. Shahir, Z. Humayun, M. A. Tamim, S. Saha, M. G. R. Alam, and A. M. Khan. CHNNet: An Artificial Neural Network With Connected Hidden Neurons. *Preprint* (2023, revised 2025). arXiv:2305.10468.

DISTRIBUTED SYSTEMS PROJECT

LamportLedger — MSc course project, BRAC University
CSE707, Distributed Computing Systems

Built a distributed ledger prototype with Rafiad Sadat Shahir and Abdullah Khondoker. The system uses Lamport logical clocks to order events across nodes.

CRYPTOGRAPHY PROJECT

SecurePass — MSc course project, BRAC University

CSE722, Applied Cryptography and Cryptographic Protocols (Spring 2026)

Co-developed a Flask service for passwordless sign-in with WebAuthn passkeys. Tested sign-in on desktop and mobile browsers and verified that the server rejected altered signatures.

ACADEMIC APPOINTMENTS AND TEACHING

Lecturer, BRAC University — May 2024 – present

- Automata and Computability (CSE331)
- Operating Systems (CSE321)
- Algorithm Analysis and Design (CSE221)

Adjunct Lecturer, BRAC University — Jan 2024 – Apr 2024

Discrete Mathematics and labs in operating systems, computer security, and object-oriented programming.

Teaching Assistant, BRAC University — May 2022 – Dec 2023

Held consultations and graded lab work and written assignments in Data Structures and Algorithms.

TECHNICAL AUTHORSHIP

Technical Author and Expert, Hyperskill / JetBrains Academy — May 2022 – 2024

Wrote lessons on machine learning, data science, cybersecurity, operating systems, and GitHub Actions. Contributed to the Python and Node.js tracks.